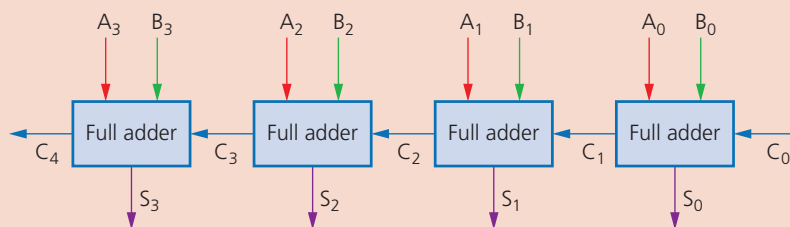


Truth table for the full adder circuit:

INPUT			OUTPUT	
A	B	C _{in}	S	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

As with the half adder circuits, different logic gates can be used to produce the full adder circuit.

The full adder is the basic building block for multiple binary additions. For example, the following diagram shows how two 4-bit numbers can be summed using four full adder circuits:



▲ Figure 10.42

In this chapter, you have learnt about:

- ✓ recognise the functions, symbols and truth tables for the logic gates: NOT, AND, OR, NAND, NOR and XOR
- ✓ create logic circuits from a given problem, logic expression or truth table
- ✓ complete a truth table from a given problem, logic (or Boolean) expression or logic circuit
- ✓ write a logic (or Boolean) expression from a given problem, logic circuit or truth table

Key terms used throughout this chapter

logic gate – an electronic circuit that relies on 'on/off' logic; the most common gates are NOT, AND, OR, NAND, NOR and XOR

logic circuit – these are formed from a combination of logic gates and designed to carry out a particular task; the output from a logic circuit will be 0 or 1

truth table – a method of checking the output from a logic circuit; a truth table lists all the possible binary input combinations and their associated outputs; the number of outputs will depend on the number of inputs; for example, two inputs have 2^2 (4) possible binary combinations, three inputs have 2^3 (8) possible binary combinations, and so on

Boolean algebra – a form of algebra linked to logic circuits and based on TRUE or FALSE